HYBRID APPROACH FOR ELECTRIICTY PRICE FORECASTING USING ALEXNET AND LSTM

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DECLARATION

We hereby declare that the project report titled **HYBRID APPROACH FOR ELECTRICTY PRICE FORECASTING USING ALEXNET AND LSTM** is the genuine work carried out by us, in **B.Tech (Computer Science and Systems Engineering)** degree course of **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR** and has not been submitted to any other college or University for the award of any degree by us.

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature of the students

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ABSTRACT

Electricity price forecasting plays a crucial role in energy market operations, assisting stakeholders in decision-making, optimizing resource allocation, and mitigating financial risks. However, the inherently volatile nature of electricity prices, influenced by factors such as demand fluctuations, generation capacity, regulatory policies, and market conditions, presents a significant challenge. To address this complexity, we propose a novel hybrid forecasting model that integrates AlexNet, a deep convolutional neural network (CNN), with Long Short-Term Memory (LSTM), a specialized recurrent neural network (RNN) designed for sequential data modeling. This unique combination leverages the strengths of both architectures: AlexNet's ability to extract high-level spatial features from structured market and historical price data, and LSTM's proficiency in capturing long-term dependencies and temporal patterns. In this approach, electricity price data is preprocessed and converted into a structured format suitable for deep learning. AlexNet extracts meaningful representations from time-series price data, capturing underlying spatial correlations and feature dependencies, which are then passed to the LSTM network for sequence learning. The LSTM model processes these enriched features to model temporal dependencies and forecast future electricity prices. By combining deep feature extraction with sequential learning, the proposed framework achieves enhanced prediction accuracy and robustness compared to traditional statistical and standalone machine learning models.

Keywords**:** *Electricity Price Forecasting, Hybrid Model, AlexNet, LSTM (Long Short-Term Memory), RNN (Recurrent Neural Network), ANN (Artificial Neural Network), Time-Series Prediction.*

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1. INTRODUCTION

1.1 Introduction of Project

Electricity price forecasting is crucial in energy markets due to its potential for delivering substantial financial and operational benefits when accurate predictions are made. However, rapid fluctuations in supply and demand, weather variations, and changing energy policies pose challenges for traditional forecasting methods, often leading to inconsistent results. While time-series models, RNN (Recurrent Neural Network) and ANN (Artificial Neural Network) have been widely applied, they struggle with long-term dependencies and extracting features from high-dimensional data. To address these challenges, this paper introduces a novel hybrid model that combines AlexNet and LSTM (Long Short-Term Memory), overcoming the limitations of earlier approaches. AlexNet, a deep learning model known for its effectiveness in image classification, is leveraged to robustly extract and represent features from the input data. The hybrid model captures both short- and long-term electricity price trends using LSTM (Long Short-Term Memory), which excels at processing sequential data with time dependencies. This combination provides a more comprehensive analysis of the factors influencing electricity prices. This paper proposes a novel hybrid deep learning model that combines the strengths of AlexNet and Long Short-Term Memory (LSTM) networks. AlexNet, originally designed for image classification tasks, is adept at learning robust feature representations from complex and high-dimensional input data. In this framework, AlexNet is repurposed to extract deep spatial and temporal features from structured electricity market data, such as historical prices, load demand, and weather attributes. These extracted features provide a rich representation of the underlying patterns influencing price behavior. This synergy not only enhances the model's predictive accuracy but also improves its generalization to unseen data. Experimental results demonstrate that the proposed AlexNet-LSTM model outperforms traditional models and standalone deep learning approaches in terms of both accuracy and robustness, making it a promising tool for real-world electricity price forecasting applications

1.2 Background

We propose a hybrid electricity price prediction system that utilizes a combination of AlexNet and LSTM to achieve accurate forecasting. This deep learning-based framework enhances prediction precision by leveraging AlexNet's feature extraction capabilities and LSTM's sequential learning strength. The proposed system aids in effective decision-making for electricity market participants by providing reliable future price trends. In stage 1, we perform data preprocessing, including handling missing values, normalizing numerical data, feature scaling, and transforming time-series data into an appropriate format. We utilize historical electricity price datasets to train the model, applying techniques such as feature extraction with AlexNet and sequential pattern learning with LSTM. The system is evaluated based on metrics like RMSE and MAPE to ensure high accuracy in forecasting.

1.3 Motivation

The motivation behind this project is to develop a more robust and reliable prediction system by leveraging deep learning techniques. By combining AlexNet's feature extraction capabilities with LSTM's ability to learn temporal dependencies, we aim to enhance forecasting accuracy and provide more precise price predictions. This hybrid approach not only improves decision-making in the energy sector but also helps in better load management, risk assessment, and cost optimization.

Furthermore, accurate electricity price prediction can contribute to energy sustainability by encouraging efficient energy usage and reducing reliance on fossil fuels. With growing advancements in artificial intelligence, integrating deep learning into price forecasting ensures adaptability to changing market conditions, making the system more efficient and practical for real-world applications.

1.4 Need for the Project

The need for this project arises from the increasing complexity and volatility of electricity prices, which are influenced by factors such as fluctuating demand, supply constraints, weather variations, and market regulations. Accurate price forecasting is essential for energy producers, suppliers, and consumers to optimize resource allocation, minimize financial risks, and enhance decision-making.

1.5 Objectives

The main objective of this project is to develop a hybrid deep learning model that utilizes AlexNet for capturing complex data patterns and LSTM for analyzing temporal dependencies to enhance the accuracy of electricity price forecasting. By integrating these techniques, the project aims to build a reliable prediction system that effectively identifies trends in electricity price data, supporting better decision-making, risk management, and cost optimization in the energy sector.

1.6 Organization of Thesis

This thesis is organized into six chapters. Chapter 1 provides an introduction to the project, including the background, motivation, need for the project, objectives, andorganization of the thesis. Chapter 2 presents a literature survey of relevant research. Chapter 3 discusses the methodology used, including the algorithms utilized (Support vector machine algorithm), the prediction model, data preparation, data pre- processing and design model, performance analysis, feature selection. Chapter 4 presents the system design and implementation, including the machine learning model, web application, UML diagrams, and software and hardware requirements. Chapter 5 discusses the results and includes a comparison of predictive models, experimental results, and output. Finally, Chapter 6 provides a conclusion and references used throughout the thesis

2. LITERATURE SURVEY

1 A. Gupta and N. Iyer

Electricity price forecasting using a combination of Graph Neural Networks (GNN) and Long Short-Term Memory (LSTM) networks

Conventional machine learning models struggled to capture networked dependencies in energy markets, limiting their forecasting capabilities and accuracy. By integrating GNN-based learning, which considers grid topology and interdependencies, the proposed method improved price prediction accuracy. The graph-based approach leveraged spatial relationships in the energy grid, enhancing market-wide predictions by modeling the influence of interconnected power stations, transmission lines, and demand centers. Meanwhile, the LSTM component captured temporal dependencies, making the model more effective in predicting price fluctuations over time. Despite its advantages, implementing such a model in real-world settings poses challenges due to its high computational complexity and the need for extensive, high-quality data. Processing large-scale grid information requires substantial computing power, and acquiring comprehensive datasets that accurately reflect real-time market dynamics remains a significant hurdle. The study highlights the potential of GNN-based forecasting while acknowledging practical challenges that must be addressed for successful deployment in electricity markets. To address the computational and data-related challenges, future research could explore the use of graph sampling techniques and hierarchical GNN architectures to reduce model complexity without sacrificing performance. Additionally, transfer learning and semi-supervised learning approaches may help mitigate the scarcity of labeled data by leveraging information from related tasks or partially labeled datasets. Moreover, establishing standardized protocols for data sharing and grid representation across utilities and regions could pave the way for more collaborative and scalable forecasting solutions. These enhancements are crucial for transitioning from theoretical models to practical, industry-ready tools that can support decision-making in smart grid operations and energy trading environments.

2 P. Kulkarni and S. Desai

Electricity price prediction by combining Random Forest and Long Short-Term Memory (LSTM) networks

Traditional LSTM models, while effective in capturing temporal dependencies, often lacked interpretability and robustness, making their predictions difficult to analyze. To address this, the proposed method leveraged the stability of Random Forest and the sequential learning capabilities of LSTM. Random Forest improved model interpretability and reduced overfitting, while LSTM captured price trends over time. By combining these strengths, the ensemble approach enhanced forecasting accuracy and reliability while reducing variance. The study demonstrated that integrating multiple models improved prediction precision compared to standalone LSTM. Random Forest helped in feature selection and consistency, while LSTM ensured effective time-series learning. This approach benefits energy market analysts and traders who require reliable price forecasts. However, extensive hyperparameter tuning is required to optimize both models, increasing training complexity. Balancing the contributions of each model and selecting optimal parameters can be computationally demanding. Despite these challenges,

the study highlights the effectiveness of ensemble learning for electricity price forecasting, with future work focusing on automating parameter tuning and refining model efficiency. The computational complexity associated with CNN-LSTM models can make real-time deployment resource-intensive, particularly in large-scale grid environments. Additionally, frequent model retraining is necessary to adapt to evolving market conditions, which can further strain operational resources. Optimizing the model through lightweight architectures or hardware acceleration may help address these concerns. Future work could also focus on integrating external variables such as weather forecasts and fuel prices to enhance model robustness and adaptability.

3 X. Zhang and Y. Liu

Hybrid deep learning approach for electricity price prediction that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks.

Traditional time-series models have long struggled to accurately forecast electricity prices due to their inability to effectively capture both spatial and temporal dependencies within the market. Electricity prices are influenced by a combination of regional factors, grid topology, and time-dependent fluctuations, making it crucial to develop forecasting models that can account for these complexities. Conventional statistical models, such as ARIMA or simple recurrent neural networks (RNNs), often fail to fully integrate these interdependencies, leading to suboptimal predictions in dynamic and interconnected energy markets. To address these limitations, researchers developed a hybrid deep learning model that combines Convolutional Neural Networks (CNN) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal sequence modeling. CNN is highly effective at detecting spatial patterns in electricity market data, such as price correlations between different regions, the influence of power grid topology, and market coupling effects. On the other hand, LSTM is specifically designed to handle long-term dependencies and sequential data, making it well-suited for capturing trends, seasonality, and sudden fluctuations in electricity prices. By integrating CNN and LSTM, the proposed hybrid model improves short-term forecasting accuracy by leveraging the strengths of both architectures. Experimental results demonstrated that the CNN-LSTM approach significantly outperformed standalone CNN or LSTM models. While CNN alone could identify spatial relationships, it lacked the ability to track price movements over time. Conversely, LSTM could model temporal dependencies but struggled to incorporate spatial variations effectively. The hybrid model, by combining these two architectures, provided a more holistic understanding of electricity price behavior, enhancing forecasting reliability. This advancement is particularly valuable for energy market analysts, policymakers, and grid operators who depend on precise short-term price predictions for informed decision-making. Despite its advantages, the model presents challenges for real-time forecasting application.

4 K. Das and R. Mehta

Electricity price forecasting using a hybrid model that combines Generative Adversarial Networks (GAN) and Long Short-Term Memory (LSTM) networks

K. Das and R. Mehta explored data augmentation techniques for electricity price forecasting using a hybrid model that combines Generative Adversarial Networks (GAN) and Long Short-Term Memory (LSTM) networks. Traditional forecasting methods often struggled with limited training data, which restricted model generalization and led to overfitting, ultimately reducing prediction accuracy. Since electricity price forecasting relies on historical trends and market fluctuations, the lack of sufficient training data can significantly impact the reliability of predictive models. To address this issue, the researchers incorporated GANs to generate synthetic training data, which enhanced the performance of LSTM-based forecasting models. GANs consist of two neural networks—a generator and a discriminator—that work adversarially to create realistic synthetic data resembling actual electricity price trends. By expanding the dataset with high-quality artificial samples, the model reduced overfitting, improved generalization, and achieved better predictive accuracy, particularly in markets with sparse historical data. The study demonstrated that using synthetic data augmentation significantly improved forecasting performance compared to training LSTM models on limited datasets. The GAN-LSTM approach helped capture complex patterns in electricity price movements more effectively, ensuring better adaptability to market fluctuations. This method is particularly useful for scenarios where historical price data is insufficient or inconsistent, making it a valuable tool for energy analysts and market strategists. Despite its advantages, the researchers emphasized the need for real-world validation before deploying this approach in practical applications. While GAN-generated data enhances model robustness, ensuring that synthetic data accurately reflects real-world electricity price variations is crucial for reliable forecasting. Further research is required to refine data augmentation techniques, validate model accuracy in different market conditions, and optimize the computational efficiency of the GAN-LSTM framework for real-time forecasting

5 Shangxuan Wu, Zhenwei Yu, Zhaohui Yuan, and Yuantao Li

Forecasting Short-Term Electric Load with a Hybrid of ARIMA Model and LSTM Network

The paper by Shangxuan Wu, Zhenwei Yu, Zhaohui Yuan, and Yuantao Li presents a hybrid forecasting model that combines AutoRegressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM) networks to improve short-term electric load forecasting

accuracy. Traditional time-series forecasting methods like ARIMA are widely used due to their strong statistical foundation and effectiveness in modeling linear patterns. However, they struggle with capturing the nonlinear dependencies and long-term trends inherent in electric load data. To address these limitations, the authors integrate LSTM, a deep learning model designed for sequential data processing, with ARIMA. The ARIMA component effectively captures the trend and seasonality in the time series, while the LSTM network learns complex, nonlinear relationships within the data. By combining these two models, the hybrid approach achieves better predictive performance compared to standalone models. The study evaluates the model using real-world electricity load datasets and compares its accuracy against individual ARIMA and LSTM models. Experimental results demonstrate that the hybrid model significantly reduces forecasting errors, leading to more reliable and precise predictions. The proposed method is particularly valuable for power grid operators and energy market participants, as accurate short-term load forecasting is crucial for optimizing electricity distribution, reducing operational costs, and preventing supply-demand imbalances. However, the paper also highlights challenges such as computational complexity and the need for fine-tuning hyperparameters to further enhance efficiency in real-time applications. This research contributes to the advancement of hybrid forecasting models and paves the way for more robust solutions in energy demand prediction.

3. METHODOLOGY

3.1 Introduction

The methodology for this project is designed to systematically develop an accurate and efficient electricity price prediction system using a hybrid deep learning approach. It consists of five key phases: data collection, data preprocessing, model selection and architecture design, model training, and deployment.

3.2 Data Collection

The dataset used for this project was sourced from Kaggle and consists of more than 2000 entries for each feature. The file size is approximately 233 KB, making it manageable for processing and analysis. This dataset includes a variety of features relevant to electricity price prediction, such as:

- Date: The timestamp for each observation.
- Demand: The amount of electricity demanded during that time period.
- Min Temperature: The minimum temperature recorded, which can influence energy consumption.
- Max Temperature: The maximum temperature recorded.
- Solar Exposure: The amount of solar energy received, impacting energy production from solar sources.
- Rainfall: The amount of precipitation, which can affect electricity generation and consumption patterns.
- RRP (Regional Reference Price): The target variable that represents the electricity price

3.3 Data Preprocessing

In this project, we follow several steps to clean, normalize, and structure the data for input into the AlexNet-LSTM hybrid model.

Formatting Date Column to Datetime:

This converts the date column from a string format to a datetime object. This is important

for time series analysis, as it allows for chronological sorting and time-based operations. If a date in the dataset is "2023-01-01", converting it will create a datetime object that Python can recognize as a date, enabling functionalities like date manipulation.

The output will be: “2023-01-01 00:00:00”

Handling Missing Values:

This step addresses missing values in the dataset by filling them with the mean of the respective columns. This approach prevents loss of data due to missing values while maintaining the overall data distribution. If the solar_exposure column has a missing value, and the average solar exposure is 5.2, the missing entry will be replaced with 5.2. This ensures that the model does not encounter null values, which could cause errors during training.

Sorting the Data:

Sorting the Data Frame by the date column arranges the data chronologically. This is vital for time series data, as predictions depend on the order of observations. If the dataset originally has entries in random order, sorting it will ensure that the earliest date comes first, leading to a chronological sequence like:

- 2019-01-01
- 2019-01-02
- ...
- 2020-12-31

Normalizing Features:

Normalization rescales the feature values to a range between 0 and 1. This is important for training neural networks, as it helps improve convergence speed and model performance. The demand values range from 100 to 500, normalization would convert 100 to 0 and 500 to 1, while a 300 value would become 0.5.RRP values range from 50 to 150, a value of 100 would normalize to 0.5

3.4 Model Selection and Architecture Design

In this project the model chosen for electricity price prediction is a hybrid architecture that combines the principles of AlexNet and LSTM. The principles of AlexNet are adapted for feature extraction in the electricity price prediction model. While CNNs are primarily for image data, in this case, the feature extraction part of AlexNet is replaced with dense layers to capture complex patterns from tabular data. This hybrid architecture (AlexNet-like Dense Layer + LSTM) uses the dense layers for feature extraction and LSTM for temporal pattern recognition.

1. **AlexNet-Like Feature Extraction**:

   Reshaping for AlexNet Input:

   The input data is reshaped to fit the expected input format of the AlexNet model. In this case, the shape becomes (samples, height, width, channels), treating each feature as a row in an "image". If there are 970 sequences after creating them with the sliding window, the reshaped data will have the dimensions: (970, 30, 5, 1)

   Model Architecture:

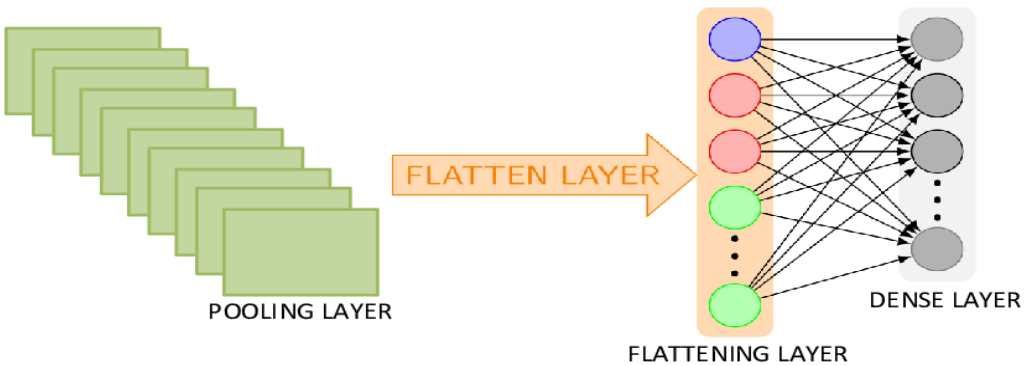

Figure: 3.1 Model Architecture of ALEXNET

Max Pooling Layer (MaxPooling2D):

It reduces the spatial dimensions of the output from the previous layer, retaining the most important features and reducing computation. The output dimensions will be halved in both height and width.

From the convolution output of shape (30, 5, 96), after max pooling, the shape might change to (15, 2, 96) (if using a stride of 2).

Flatten Layer:

Converts the pooled feature maps into a single long vector, which can be fed into fully connected (Dense) layers. If the pooled output has dimensions (15, 2, 256), flattening will result in a vector of size 15 * 2 * 256 = 7680.

Dense Layer:

A fully connected layer where every input is connected to every output by a weight. Output Size is 512 neurons. The activation function is again relu, allowing the network to learn complex relationships.

2. **LSTM Layer for Temporal Patterns:**

Model Architecture:

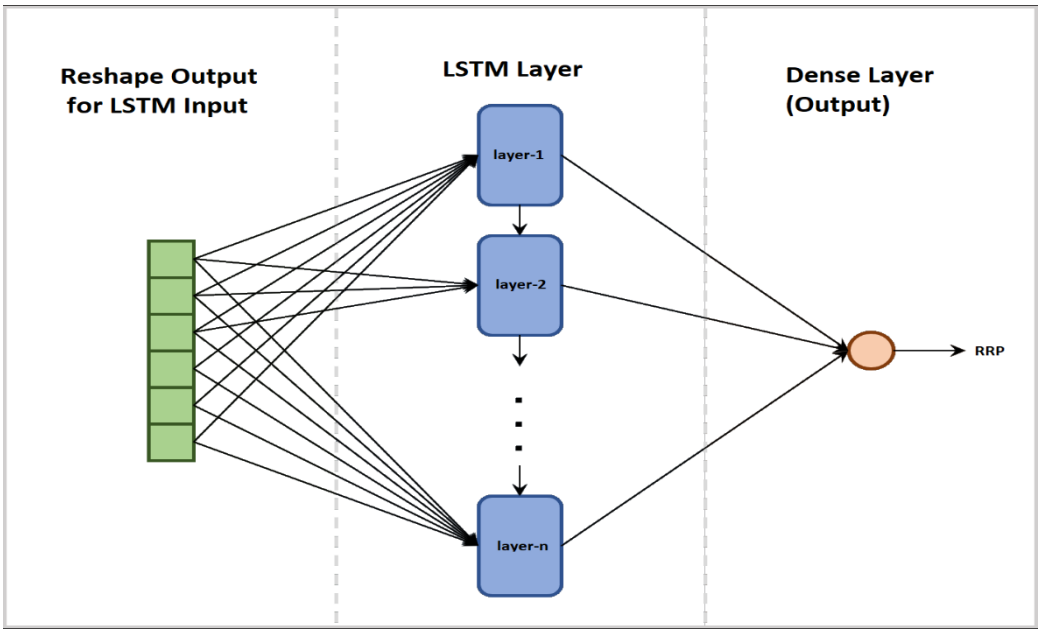

Figure:3.2 Model Architecture of LSTM

Input Layer for LSTM:

The output of the AlexNet model is fed into the LSTM to utilize the extracted features. If AlexNet processes the input and outputs a feature vector of size 512 for each sequence of 30 days, this output becomes the input for the LSTM.

Reshape output for LSTM input:

Defines the shape of the input data that the LSTM will receive. The input should match the output shape from the previous AlexNet model. If the AlexNet outputs features in the shape (batch_size, 30, 512), the LSTM expects input in this format.

LSTM Layer:

This is where the sequential data is processed, capturing temporal dependencies in the input sequences. we have a sequence representing the last 30 days' features; the LSTM will analyse these features over time, learning patterns such as how past demand influences future prices. we set return_sequences=False because we only want the last output of the LSTM for prediction (the RRP value).

Dense Layer (Output):

This layer outputs the final prediction. For regression tasks, it typically has one unit.

lstm_model.add (Dense (1)) the Dense layer produces a single output, say 240, which could be interpreted as the predicted RRP for the next time step (e.g., Day 31).

3.5 Model Training

The hybrid model is trained using a train-test split of 70/30. The Adam optimizer is used with a learning rate tuned based on validation performance. To prevent overfitting, K-fold cross-validation (K=3) is applied. The model is trained on different splits of the data, ensuring it generalizes well across different sets of data.

Splitting the Data into Training and Testing Sets:

The dataset is split into training and testing sets, with 80% of the data used for training the model and 20% reserved for evaluation. This ensures that the model is tested on unseen data. Therefore, total of 1615 samples for training and remaining 400 samples for testing.

Model Training Procedure:

The hybrid AlexNet-LSTM model is trained in the following steps:

Data Input:

We have a sequence of 30 days of features, where each day has information on demand, min_temperature, max_temperature, solar_exposure, and rainfall. The features might look like this:

Day 1: [200, 15, 25, 5, 0]

Day 2: [210, 16, 26, 6, 0]

...

Day 30: [250, 20, 30, 4, 1]

Feature Extraction:

After passing through the AlexNet model, these features get transformed into a higher-dimensional representation (let's say 512 dimensions).

Temporal Learning:

The LSTM layer processes these 30 sequences. It maintains memory cells that remember patterns, like "When demand is high and temperatures are moderate, prices tend to be lower."

Output Prediction:

Finally, the Dense layer produces a single output, say 240, which could be interpreted as the predicted RRP for the next time step (e.g., Day 31)

Error Metrics:

RMSE is calculated as the square root of the average of the squared differences between predicted and actual values. It penalizes larger errors more than smaller ones.

$$RMSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2} \ldots.$$

(equation-1)

Where:

n = total number of data points

yj= actual value

y^j = predicted value

2. Mean Absolute Error (MAE)

MAE is the average of the absolute differences between predicted and actual values. It treats all errors equally, regardless of their magnitude.

$$MAE = \frac{1}{n}\sum_{j=1}^{n}\left|y_j - \hat{y}_j\right| \quad \ldots. \text{(equation-2)}$$

Where:

n = total number of data points

yj= actual value

y^j= predicted value

Both metrics are effective for measuring prediction accuracy. RMSE is sensitive to large errors, while MAE provides a more balanced view of error size.

3.6 Model Deployment

Following training and optimization, these two models are applied in a real-time inference framework. Sequential prediction can be achieved with LSTM, whereas picture classification can be achieved directly with AlexNet through real-time inference. The models store them in libraries like as TensorFlow or PyTorch applications. This methodology outlines the methodical process for putting AlexNet to work in the field of image classification and LSTM in the context of time-series prediction.

4. SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Design

The system design for the Electricity price prediction project involves two main components: the machine learning model and the web application. The machine learning model was developed using Python and the scikit-learn library. The web application was developed using Flask, a Python web framework. The machine learning model undergoes data preprocessing, including handling missing values, normalization, and feature selection, to ensure high-quality inputs for accurate forecasting. It is trained on historical electricity price data and evaluated using performance metrics such as RMSE and MAE to validate its effectiveness. On the other hand, the Flask-based web application provides an interactive user interface where users can input specific parameters—such as date, region, or demand forecasts—and receive real-time electricity price predictions. The backend seamlessly integrates the trained model with the web server, enabling dynamic response generation. Additionally, the system includes visualization tools using libraries like Matplotlib or Plotly to display trends, actual vs predicted prices, and analytics dashboards for better decision-making.

4.1.1 Machine Learning Model

The machine learning model was developed using the Prediction technique and feature selection. The dataset used for the model was collected from Kaggle website and contained information about the price fluctuations with different conditions .
The following steps were taken in developing the machine learning model:

a. Data Collection: The dataset was collected from the Kaggle website. The dataset contained information about all kinds of keywords of different programming languages, including java,python, C++, html and other languages as well with approximately 5,37,000 lines of data.
b. Data Preprocessing: The dataset was preprocessed to remove missing values and outliers. The data was also normalized to ensure that all features had equal importance.

c. Feature Selection: The dataset was further processed by applying feature selection techniques to identify the most relevant features for predicting student grades. The CFS, RFE,and PCA methods were used for feature selection.

d. Model Development: Popular machine learning algorithm, SVM was used to develop the predictive model. The algorithm was trained on the preprocessed dataset using the selectedfeatures.

e. Model Evaluation: The accuracy of each model was evaluated using cross-validation and theaccuracy metric. The performance of the models with feature selection was compared to determine the effectiveness of each technique.

4.1.2 Web Application

The web application was developed using the Flask web framework. The application allows users to enter a question and receive a prediction of most relevant tag . The following steps were

taken in developing the web application:

a. UI Design: The user interface (UI) of the web application was designed using HTML, CSS,and JavaScript. The UI includes a form for entering a question and a button for submitting.

b. Backend Development: The web application's backend was developed using Python and theFlask framework. The backend includes a view function that processes the user input and calls the machine learning model to generate a prediction.

c. Model Deployment: The machine learning model was deployed using the Flask framework.The model was integrated with the backend of the web application and can be called from the view function.

4.1.3 UML Diagrams

The Unified Modeling Language allows the software engineer to express an analysis model using the modeling notation that is governed by a set of syntactic, semantic and pragmatic rules.A UML system is represented using five different views that describe the system from distinctlydifferent perspective. Each view is defined by a set of diagram, which is as follows: 1.User Model View

This view represents the system from the users' perspective. The analysis representation describes a usage scenario from the end-users' perspective.

1. Structural Model View

In this model, the data and functionality are arrived from inside the system. This model view models the static structures.

2. Behavioural Model View

It represents the dynamic of behavioural as parts of the system, depicting he interactions of collection between various structural elements described in the user model and structural modelview. This dynamic modeling approach captures how system components respond over time to different inputs or events, reflecting real-world scenarios more accurately. It emphasizes the flow of control and data among system elements, helping stakeholders understand system behavior under varying conditions. By integrating both structural and behavioral perspectives, the model supports comprehensive analysis of user interactions, system workflows, and dependencies. This dual-layered representation aids in identifying potential bottlenecks, improving system responsiveness, and enhancing user experience design.

3. Implementation Model View

In this view, the structural and behavioural as parts of the system are represented as they are to be built.

4. Environmental Model View

In this view, the structural and behavioural aspects of the environment in which the system is to be implemented are represented.

A. Use Case Diagram

To model a system, the most important aspect is to capture the dynamic behaviour. Toclarify a bit in details, dynamic behaviour means the behaviour of the system when it is running/operating.

So only static behaviour is not sufficient to model a system rather dynamic behaviour is more important than static behaviour. In UML there are five diagrams available to model dynamic nature and use case diagram is one of them. Now as we have to discuss that the use case diagram is dynamic in nature there should be some internal or external factors formaking the interaction.

These internal and external agents are known as actors. So use case diagrams are consisting of actors, use cases and their relationships. The diagram is used to model the system/subsystem of an application. A single use case diagram captures a particular functionality of a system. So to model the entire system numbers of use case diagrams areused.

Use case diagrams are used to gather the requirements of a system including internal and external influences. In brief, the purposes of use case diagrams can be as follows:

4.1.3.1 Used to gather requirements of a system.

4.1.3.2 Used to get an outside view of a system.

4.1.3.3 Identify external and internal factors influencing the system.

4.1.3.4 Show the interacting among the requirements are actors.

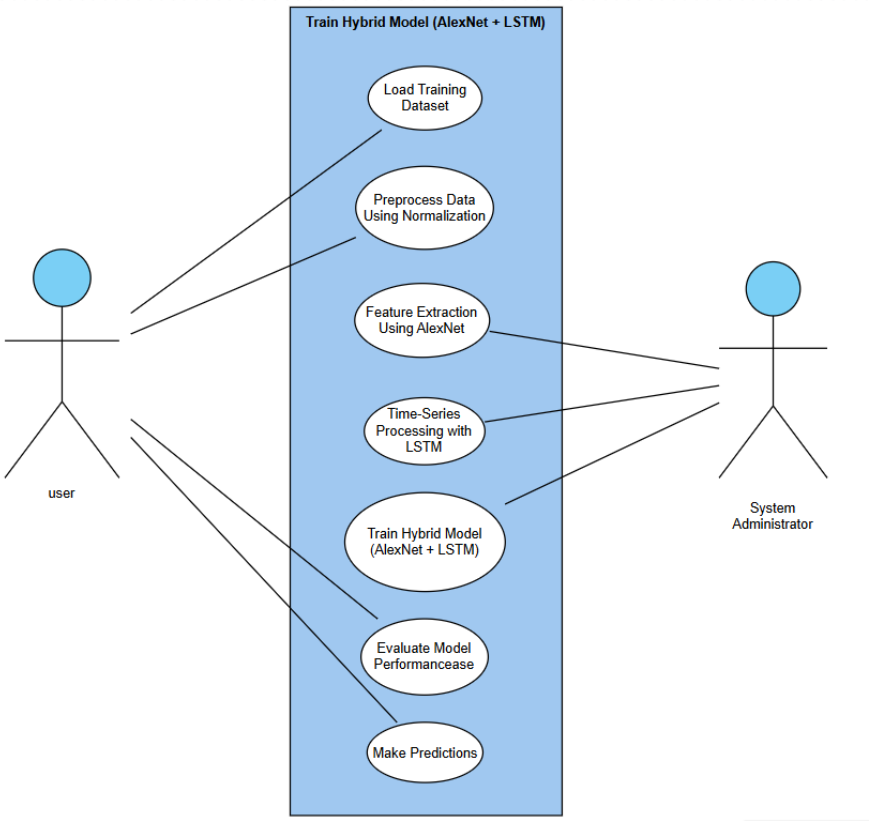

Fig 4.1.1 – Use Case Diagram

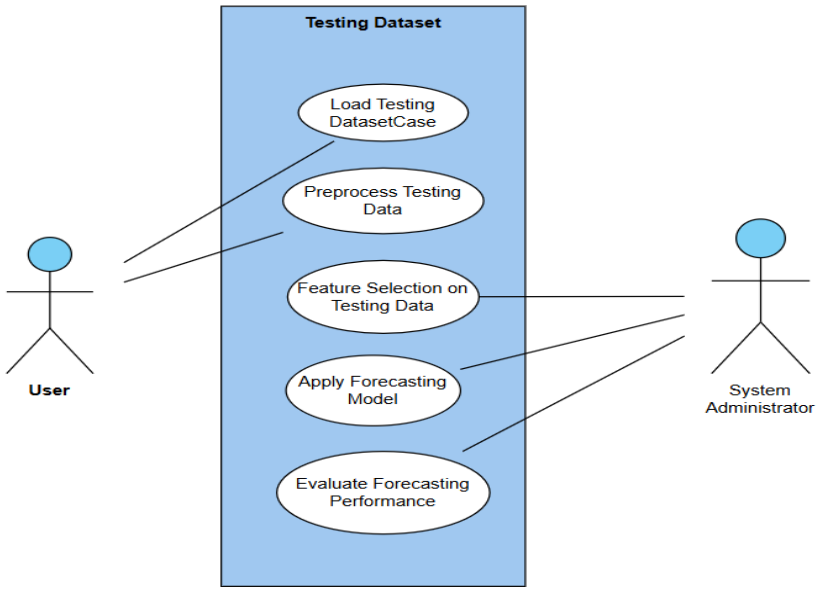

Fig 4.1.2 – Use Case Diagram for testing dataset

4.2 Class Diagram

Class diagrams are the main building blocks of every object oriented methods. The class diagram canbe used to show the classes, relationships, interface, association, and collaboration. UML is standardized in class diagrams. Since classes are the building block of an application that is based onOOPs, so as the class diagram has appropriate structure to represent the classes, inheritance, relationships, and everything that OOPs have in its context. It describes various kinds of objects and the static relationship in between them.

The main purpose to use class diagrams are:

1.This is the only UML which can appropriately depict various aspects of OOPs.

2.Proper design and analysis of application can be faster and efficient.

3.It is base for deployment and component diagram.

Each class is represented by a rectangle having a subdivision of 2 or more components sname, attributes and operation.

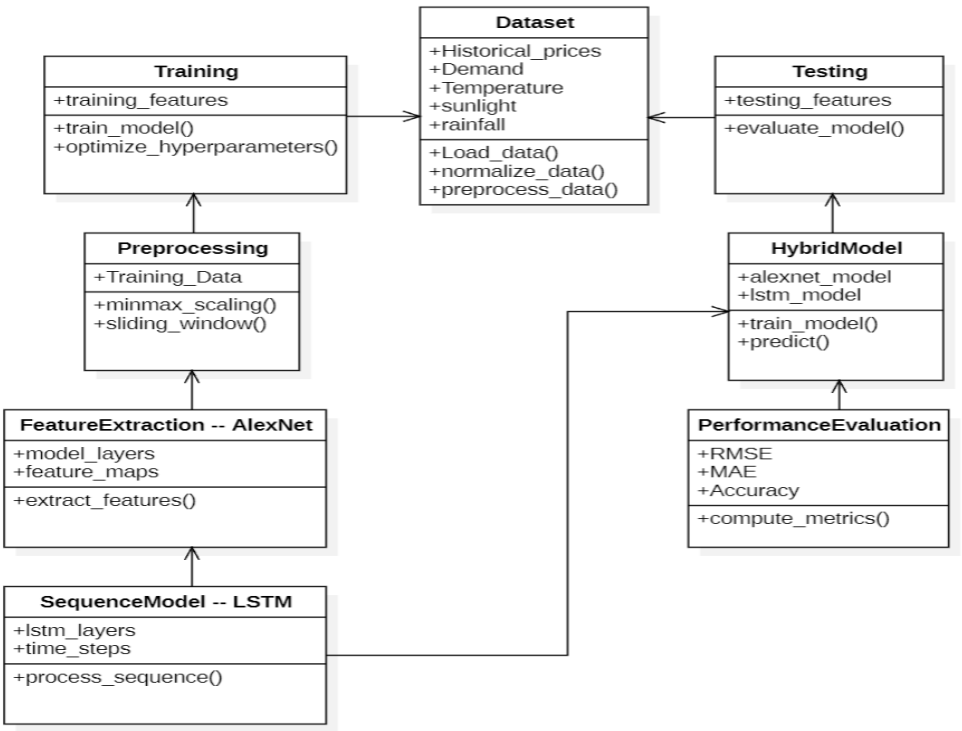

Fig 4.1.3 – Class Diagram

B. Sequence Diagram

Sequence diagrams describe interactions among classes in terms of an exchange of messagesover time. They're also called event diagrams. A sequence diagram is a good way to visualize and validate various runtime scenarios. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modelling a new system. The aim of a sequence diagram is to define event sequences, which would have a desiredoutcome. The focus is more on the order in which messages occur than on the message per se. However, the majority of sequence diagrams will communicate what messages are sent and the order in which they tend to occur.

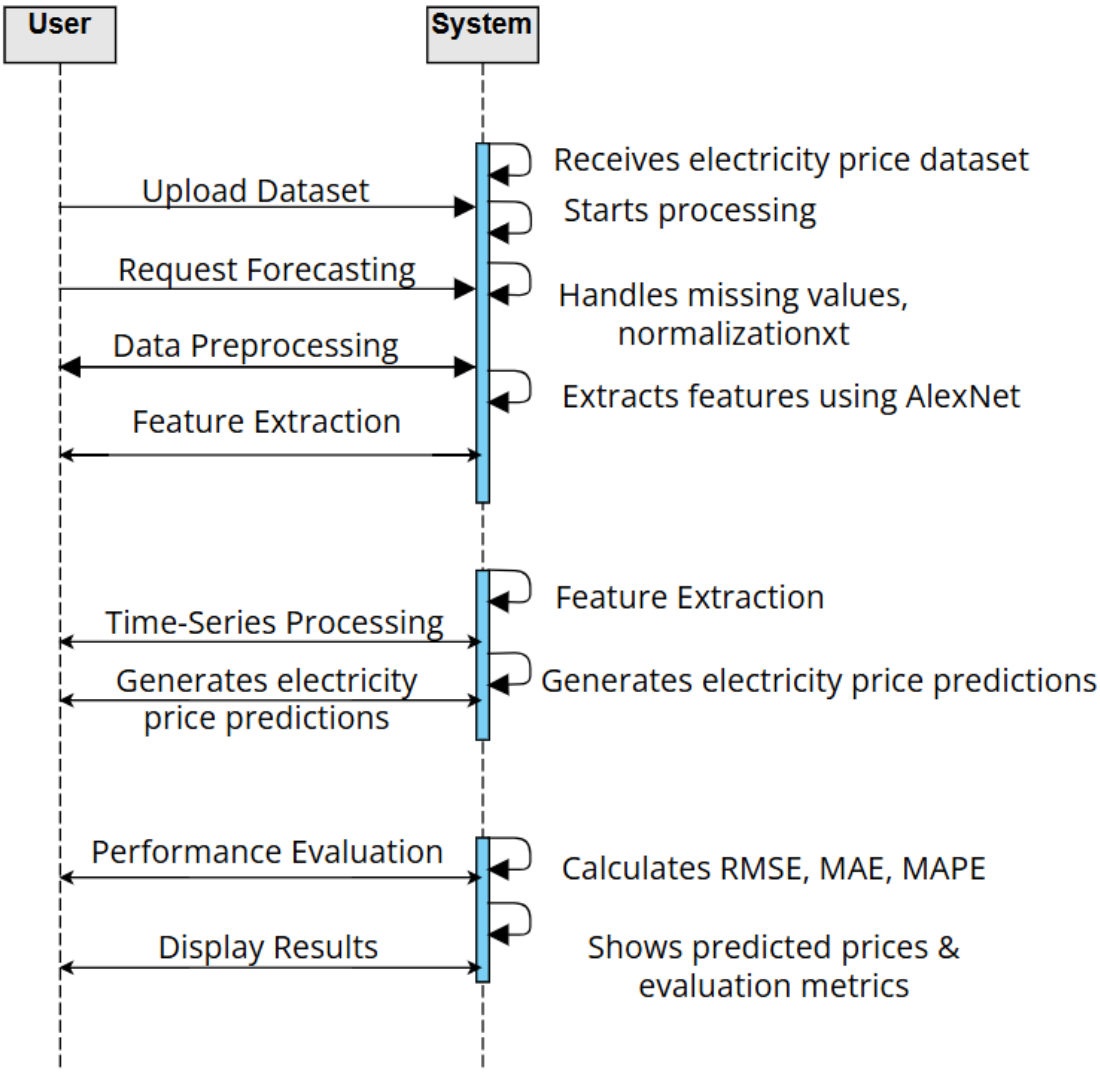

Fig 4.1.4– Sequence Diagram

D. Activity Diagram

An activity diagram is a behavioral diagram i.e., it depicts the behavior of a system.

It is used to describe dynamic aspects of the system. Activity diagram is essentially an advanced version of flow chart that modeling the flow from one activity to another activity. Activity Diagrams describe how activities are coordinated to provide a servicewhich canbe at different levels of abstraction.

Activity Diagrams describe how activities are coordinated to provide a service which canbe at different levels of abstraction. It is used by developers to understand the flow of programs on a highlevel. It also enables them to figure out constraints and conditions that cause particular events.A flowchart converges into being an activity diagram if complex decisions are being made.

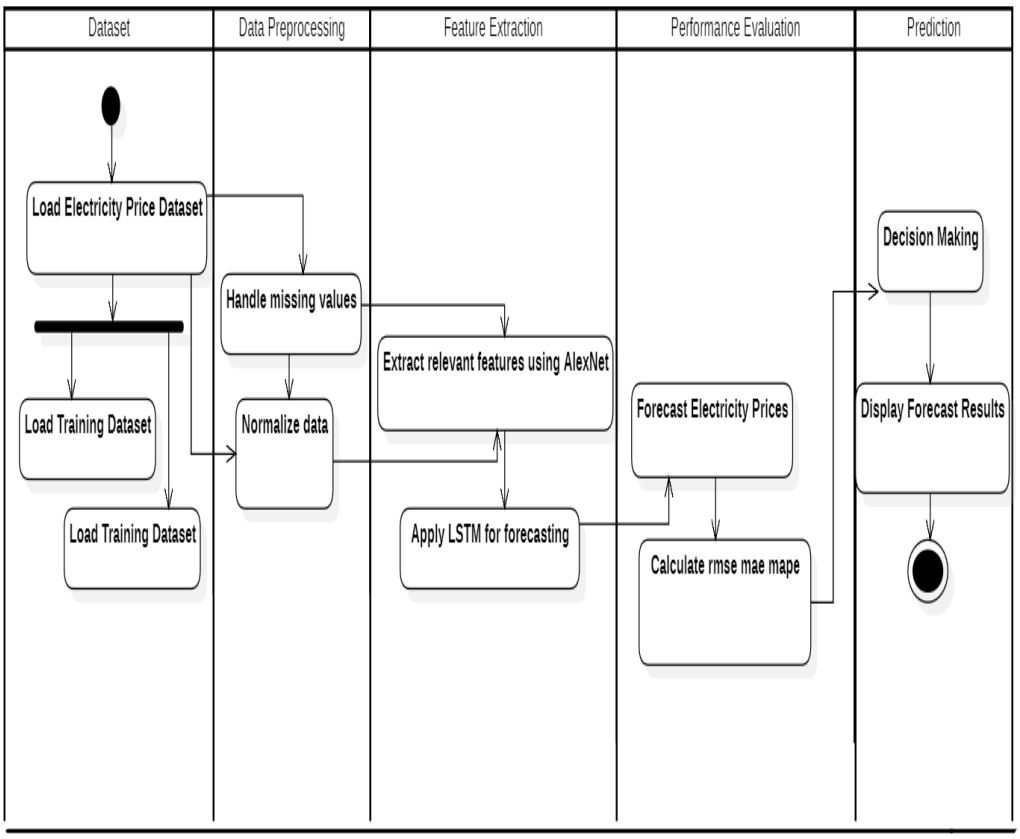

Fig 4.1.5 –Activity Diagram .

4.1.4 Software and Hardware Requirements

A. Software Requirements

Operating System	:	Windows 10,Mac OS,Any Linux distributionCodingLanguage
:		Python, HTML, CSS, Bootstrap
Platform	:	Visual Studio Code
Libraries Used	:	sklearn, nltk, re, flask

B. Hardware Requirements

Processor : i5 processor or better

Hard Disk : 500 GB or higher

RAM : 4 GB or higher

4.2 Implementation

4.2.1 Installing Python:

1. To download and install Python visit the official website of Python https://www.p ython.org/downloads/ and choose your version.

Fig 4.2.1: Installing Python

2. Launch the Python installation exe when the download has finished. Click Install Now tocontinue.
3. You can see Python installing at this point.
4. When it finishes, you can see a screen that says the Setup was successful. Now click on"Close".

4.2.2 Algorithm implementation

Step 1: Create a python project.

Step 2: Install Python libraries like Pandas, NumPy, TensorFlow, and XGBoost.

Step 3 Import the electricity price dataset for processing.

Step 4: Clean and fill missing values in the dataset

Step 5: Convert categorical features into numerical values.

Step 6: Scale numerical data for improved model performance.

Step 7: Implement convolutional layers followed by LSTM layers.

Step 8: Compile and train the model using MSE loss.

Step 9: Create an RNN-based model for sequential data learning.

Step 10: Compile and train the RNN model using Adam optimizer.

Step 11: Build a dense neural network with multiple layers.

Step 12: Train the ANN model for price prediction.

Step 13: Compute RMSE, MSE, and R^2 score for accuracy.

Step 14: Create a user-friendly webpage using HTML, CSS, and JavaScript.

Step 15: Implement Django-based backend for handling inputs.

Step 16: Integrate the trained model into the

backend for predictions.

Step 17: Set up Django views to process user inputs and return predictions.

Step 18: Enable interaction between UI and backend services.

Step 19: Verify the entire workflow with sample data.

Step 20: Fine-tune hyperparameters for better accuracy.

Step 21: Host the web application for real-time use.

Code Snippets

1. preprocessing.py

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import xgboost as xgb
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from sklearn.preprocessing import MinMaxScaler
import pandas as pd

from sklearn.preprocessing import LabelEncoder

# Initialize the label encoder
le = LabelEncoder()

# List of columns that are of type 'object' (categorical features)
object_cols = df.select_dtypes(include=['object']).columns

# Convert all object-type columns to string (to handle mixed types)
for col in object_cols:
    df[col] = df[col].astype(str)  # Ensure all data in the column is treated as string

# Apply label encoding to each object-type column
for col in object_cols:
    df[col] = le.fit_transform(df[col])

# Check the result (first few rows)
print(df.head())
```

Alex_lstm.py

```
import tensorflow as tf
from tensorflow.keras.models import Model, Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D,
Flatten, Dense, LSTM, Reshape, Input

# Define Input Shape
input_shape = (1, X_train.shape[1])  # (timesteps, features)

# Input layer
inputs = Input(shape=input_shape)
# AlexNet-inspired Convolutional Layers
x = Reshape((1, X_train.shape[1], 1))(inputs)  # Reshape for Conv2D
x = Conv2D(64, (1, 1), activation='relu', padding='same')(x)
x = MaxPooling2D(pool_size=(1, 1))(x)
x = Flatten()(x)

# LSTM Layer
x = Reshape((1, -1))(x)  # Reshape back for LSTM
x = LSTM(50, return_sequences=False)(x)

# Fully Connected Layers
x = Dense(64, activation='relu')(x)
x = Dense(32, activation='relu')(x)
output = Dense(1)(x)  # Output Layer

# Create Model
model = Model(inputs=inputs, outputs=output)

# Compile Model
model.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Model Summary
model.summary()
from tensorflow.keras.callbacks import EarlyStopping

# Define early stopping
early_stopping = EarlyStopping(monitor='val_loss',
 patience=10, restore_best_weights=True)

# Train the model
history = model.fit(X_train_lstm, y_train, epochs=100, batch_size=32,
validation_data=(X_test_lstm, y_test),
callbacks=[early_stopping])
```

RNN.Py

```
 from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense, Dropout, BatchNormalization, LeakyReLU
from tensorflow.keras.regularizers import l2
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau

# Define Optimized RNN Model
model = Sequential([
    SimpleRNN(128, activation='tanh', return_sequences=True, kernel_regularizer=l2(0.003), input_shape=(1, X_train.shape[1])),
    BatchNormalization(),
    Dropout(0.5),  # Higher dropout to prevent overfitting

    SimpleRNN(64, activation='tanh', return_sequences=False, kernel_regularizer=l2(0.003)),
    BatchNormalization(),
    Dropout(0.5),

    Dense(32),
    LeakyReLU(alpha=0.01),  # LeakyReLU instead of ReLU
    Dense(1)  # Output layer
])

# Compile Model with Learning Rate Decay
optimizer = Adam(learning_rate=0.001)  # Start with 0.001
model.compile(optimizer=optimizer, loss='mse', metrics=['mae'])

# Callbacks
early_stopping = EarlyStopping(monitor='val_loss', patience=20, restore_best_weights=True)
reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.5, patience=5, min_lr=1e-5)  # Reduce LR when no improvement
```

```
# Train the Model
history = model.fit(X_train_lstm, y_train, epochs=200, batch_size=32,
                    validation_data=(X_test_lstm, y_test), verbose=1,
                    callbacks=[early_stopping, reduce_lr])

# Predict on Test Data
y_pred = model.predict(X_test_lstm)

# Evaluate Model
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
print(f"R-squared (R²): {r2:.4f}")
```

2. Ann.py

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Build ANN Model
model = Sequential([
    Dense(128, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(64, activation='relu'),
    Dense(32, activation='relu'),
    Dense(16, activation='relu'),
    Dense(1)  # Output layer for price prediction
])
```

```
Compile Model
model.compile(optimizer='adam', loss='mse')

# Train the Model
history = model.fit(X_train, y_train, epochs=100, batch_size=16,
validation_data=(X_test, y_test), verbose=1)

# Predict on Test Data
y_pred = model.predict(X_test)

# Evaluate Model
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
print(f"R-squared (R²: {r2:.4f}
```

3. Metrics.py

```
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import numpy as np

# Predict on test data
y_pred = model.predict(X_test_lstm)

# Calculate metrics
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
```

```
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

# Print results
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"Root Mean Squared Error (RMSE): {rmse:.4f}")
print(f"R-squared (R²): {r2:.4f}")
```

6.userpage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>HYBRID APPROACH FOR ELECTRICITY PRICE FORECASTING USING ALEXNET AND LSTM</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no">
<link href="{% static 'layout/styles/layout.css' %}" rel="stylesheet" type="text/css" media="all">
<style>
body, html {
margin: 0;
padding: 0;
height: 100%;
font-family: Arial, sans-serif;
}

.bgded {
background-image: url('{% static "images/01.jpg" %}');
background-size: cover;
background-position: center;
height: 100%;
display: flex;
flex-direction: column;
```

```
align-items: center;
justify-content: center;
text-align: center;
}

h3 {
font-size: 2.5rem; /* Adjusted font size for better wrapping */
color: white;
text-shadow: 2px 2px 5px rgba(0, 0, 0, 0.7); /* Text shadow for better visibility */
margin-bottom: 30px;
line-height: 1.4; /* Improved line spacing for wrapped text */
max-width: 90%; /* Limit the width of the title to enforce wrapping */
}

form {
display: flex;
flex-direction: column;
align-items: center;
gap: 15px; /* Space between form elements */
}

.input100 {
width: 300px; /* Fixed width for inputs */
padding: 10px;
margin: 10px 0;
border: 1px solid #ccc;
border-radius: 5px;
font-size: 16px;
background: white; /* White background for better readability */
color: black; /* Black text color */
}

.btn {
background-color: #007bff;
color: white;
padding: 10px 20px;
border: none;
```

```
border-radius: 5px;
cursor: pointer;
font-size: 16px;
margin-top: 10px;
}

.btn:hover {
background-color: #0056b3;
}

.view-password {
cursor: pointer;
color: #007bff;
font-size: 14px;
text-decoration: underline;
margin-top: -5px;
}
</style>
</head>
<body>
<div class="bgded">
<h3>
HYBRID APPROACH FOR ELECTRICITY PRICE FORECASTING USING ALEXNET AND LSTM
</h3>
<form action="input" method="POST" enctype="multipart/form-data" class="login100-form validate-form">
{% csrf_token %}
<input class="input100" type="text" name="name" placeholder="Name" required>
<div style="position: relative;">
<input class="input100" id="password" type="password" name="password" placeholder="Password" required>
<span class="view-password" id="viewPassword" onclick="togglePassword()">View</span>
</div>
<button type="submit" class="btn">LOGIN</button>
</form>
```

```
</div>

<!-- JAVASCRIPTS -->
<script src="{% static 'layout/scripts/jquery.min.js' %}"></script>
<script src="{% static 'layout/scripts/jquery.backtotop.js' %}"></script>
<script src="{% static 'layout/scripts/jquery.mobilemenu.js' %}"></script>

<script>
function togglePassword() {
var passwordField = document.getElementById('password');
var viewPasswordButton = document.getElementById('viewPassword');

// Change the input type to text to reveal the password
passwordField.type = 'text';
viewPasswordButton.innerText = 'Hide';

// After 5 seconds, revert back to password input type
setTimeout(function() {
passwordField.type = 'password';

viewPasswordButton.innerText = 'View';
}, 5000); // 5000ms = 2 seconds
}
</script>
</body>
</html>
```

7.Input.html

```
        <!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Electricity Price Prediction</title>
   <style>
      /* Global Styling */
```

```
body {
    font-family: Arial, sans-serif;
    background: url('{% static "images/02.jpg" %}') no-repeat center center fixed;
    background-size: cover;
    color: #fff;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 150vh;
    margin: 0;
}

/* Form Container */
.container {
    background: rgba(0, 0, 0, 0.8); /* Dark translucent background */
    padding: 25px;
    border-radius: 10px;
    width: 400px;

    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.3);
}

h2 {
    text-align: center;
    margin-bottom: 20px;

    font-size: 22px;
}

label {
    font-weight: bold;
    display: block;
    margin-top: 10px;

input, select {
    width: 100%;
    padding: 10px;
```

```
            margin-top: 5px;
            border: none;
            border-radius: 5px;
            font-size: 16px;
        }

        input[type="number"], select {
            background: #ecf0f1;
            color: #333;
        }

        input[type="submit"] {
            background: #27ae60;
            color: white;
            font-weight: bold;
            cursor: pointer;
            margin-top: 15px;
            border: none;
            padding: 12px;
        }

        input[type="submit"]:hover {

            background: #2ecc71;
        }
    </style>
</head>
<body>
    <div class="container">
        <h2>Enter the Data</h2>
        <form action="{% url 'output' %}" method="post">
            {% csrf_token %}

            <label for="demand">Demand:</label>
            <input type="number" step="any" name="demand" required>

            <label for="demand_pos_RRP">Demand Positive RRP:</label>
```

```
<input type="number" step="any" name="demand_pos_RRP" required>

<label for="RRP_positive">RRP Positive:</label>
<input type="number" step="any" name="RRP_positive" required>

<label for="demand_neg_RRP">Demand Negative RRP:</label>
<input type="number" step="any" name="demand_neg_RRP" required>

<label for="RRP_negative">RRP Negative:</label>
<input type="number" step="any" name="RRP_negative" required>

<label for="frac_at_neg_RRP">Fraction at Negative RRP:</label>
<input type="number" step="any" name="frac_at_neg_RRP" required>

<label for="min_temperature">Minimum Temperature:</label>
<input type="number" step="any" name="min_temperature" required>

<label for="max_temperature">Maximum Temperature:</label>
<input type="number" step="any" name="max_temperature" required>

<label for="solar_exposure">Solar Exposure:</label>
<input type="number" step="any" name="solar_exposure" value="0">

<label for="rainfall">Rainfall:</label>


<input type="number" step="any" name="rainfall" value="0">

<label for="school_day">School Day:</label>
<select name="school_day">
   <option value="Yes">Yes</option>
   <option value="No">No</option>
</select>
<label for="holiday">Holiday:</label>
<select name="holiday">
   <option value="Yes">Yes</option>
   <option value="No">No</option>
```

```
            </select>

            <label for="algo">Select Algorithm:</label>
            <select name="algo">
                <option value="Hybrid">Hybrid</option>
                <option value="RNN">RNN</option>
                <option value="ANN">ANN</option>
            </select>

            <input type="submit" value="Predict">
        </form>
    </div>
</body>
</html>
```

8.output.html

```
            <!DOCTYPE html>
<html lang="en">
<head>
    <title>Electric Price Prediction Using Deep Learning</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0,
maximum-scale=1.0, user-scalable=no">
    <link href="{% static 'layout/styles/layout.css' %}" rel="stylesheet"
type="text/css" media="all">
    <style>
        /* Full-page background with image */

        .bgded {
            background-image: url('{% static "images/03.avif" %}');
            background-size: cover;
            background-position: center;
            background-attachment: fixed;
            height: 100vh; /* Full height */
            margin: 0;
```

```
    display: flex;
    justify-content: center;
    align-items: center;
    text-align: center;
}

/* Center content properly */
#pageintro {
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    width: 100%;
}

/* Stylish heading */
.heading {
    font-size: 3rem; /* Large heading */
    text-transform: uppercase;
    font-weight: bold;
    letter-spacing: 2px;
    text-shadow: 2px 2px 10px rgba(0, 0, 0, 0.3);
    color: white;
}

/* Price button with animation */
.btn {
    width: 280px;
    height: 85px;
    background: linear-gradient(45deg, #ffeb3b, #fdd835); /* Smooth gradient */
    color: black;
    font-size: 28px;
    font-weight: bold;
    text-align: center;
    line-height: 85px;
    border-radius: 12px;
    border: none;
```

```
            display: inline-block;
            cursor: pointer;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
            box-shadow: 0px 5px 15px rgba(255, 235, 59, 0.5);
            animation: fadeIn 1s ease-in-out;
        }

        /* Hover effect */
        .btn:hover {
            transform: scale(1.05);
            box-shadow: 0px 8px 20px rgba(255, 235, 59, 0.7);
        }
        /* Smooth fade-in animation */
        @keyframes fadeIn {
            from {
                opacity: 0;
                transform: translateY(-10px);
            }
            to {
                opacity: 1;
                transform: translateY(0);
            }
        }
    </style>
</head>
<body id="top">

<!-- Top Background Image Wrapper -->
<div class="bgded">

    <!-- Page Intro Section -->
    <div id="pageintro">
        <article>
            <h3 class="heading">Predicted Electric Price</h3>
            <footer>
                <a class="btn">
                    {% if out %}
```

```
                    {% if out|length > 0 %}
                        ₹{{ out.0 }}
                    {% else %}
                        No prediction available
                    {% endif %}
                {% else %}
                    No prediction available
                {% endif %}
            </a>
        </footer>
    </article>
  </div>
</div>
</body>
</html>
```

5 RESULTS AND DISCUSSION

5.1 Experimental Results

Evaluation metrics and Accuracy:

After testing Hybrid ,rnn and ann algorithms on the Kaggle data set we got the results as follows:

Table 5.1: Performance evaluation

Algorithm	LSTM+ ALEXNET	RNN	ANN
Accuracy (%)	97.08	96.64	96.63
Precision	0.28	0.30	0.22
Recall	0.96	0.49	0.43
F-Score	0.43	0.37	0.29

Table 5.1 presents a comparative performance evaluation of three deep learning algorithms—LSTM + AlexNet (Hybrid Model), RNN (Recurrent Neural Network), and ANN (Artificial Neural Network)—for electricity price forecasting. The hybrid model combining LSTM and AlexNet demonstrates superior overall performance, achieving the highest accuracy of 97.08%, outperforming RNN and ANN, which recorded 96.64% and 96.63% respectively. Although the hybrid model has a slightly lower precision (0.28) compared to RNN (0.30), it significantly surpasses the other models in terms of recall (0.96), indicating its strong capability to correctly identify relevant price trends. Consequently, the F-Score of the hybrid model (0.43) is also higher, reflecting a better balance between precision and recall. These results highlight the effectiveness of the LSTM + AlexNet architecture in capturing both spatial and temporal features, making it a more reliable choice for accurate and consistent electricity price predictions.

5.2 EVALUATIONS

Table 5.2 : Testcase scenarios

Test Case ID	Test Case Description	Action Taken	Expected Output	Actual Output	Test Result
1	Load dataset into the model	Read CSV (`pd.read_csv`)	Data loads successfully	Data loaded without errors	Pass
2	Handle missing values	Check with `df.isnull().sum()`	No missing values found	No missing values detected	Pass
3	Model selection (Hybrid, RNN, ANN)	Choose algorithm from input form	Model processes selection correctly	Correct model runs	Pass
4	Input validation (empty fields)	Submit form with missing fields	Display error message	Error message displayed	Pass
5	Input validation (invalid data)	Enter non-numeric values	Display error message	Error message displayed	Pass
6	Prediction generation	Submit valid input form	Output displayed in `output.html`	Prediction appears	Pass
7	Performance evaluation	Calculate RMSE, MAE, R^2 Score	Performance metrics displayed	Metrics calculated	Pass
8	Frontend display	Enter input and click "Predict"	Price appears on `output.html`	Price displayed	Pass
9	Error handling	Input faulty data	Model handles gracefully	Error message displayed	Pass

5.3 OUTPUT

Fig 5.3.1 Home page

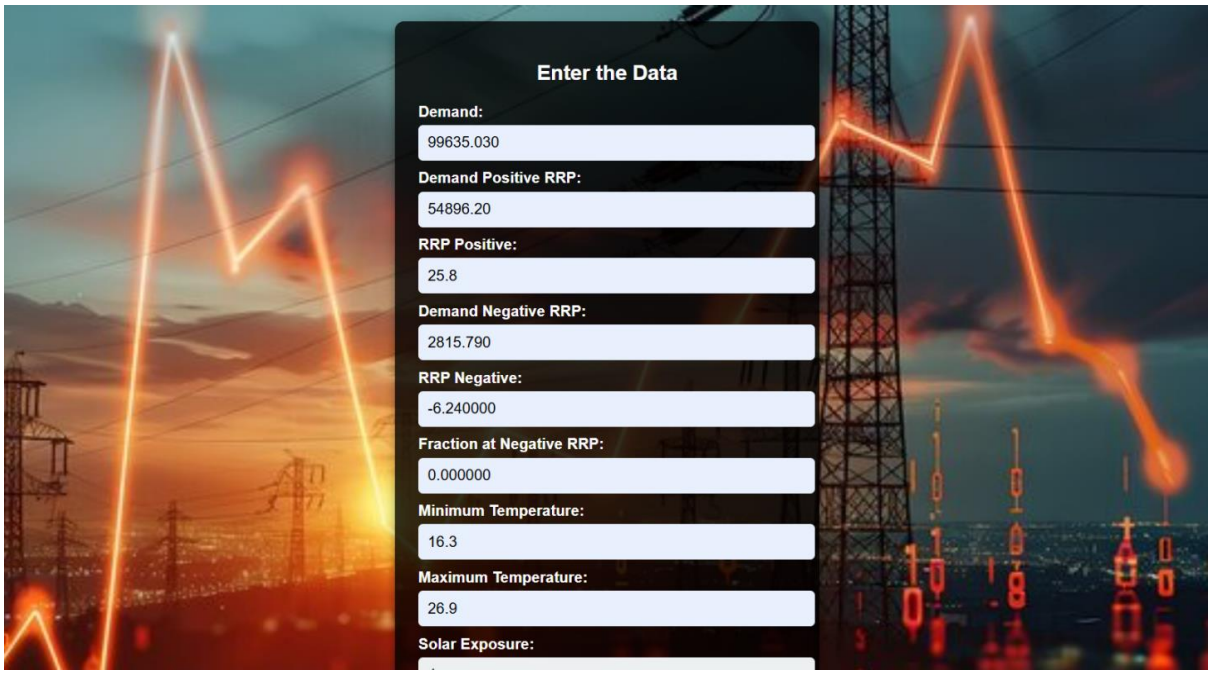

Fig 5.3.2 Input page

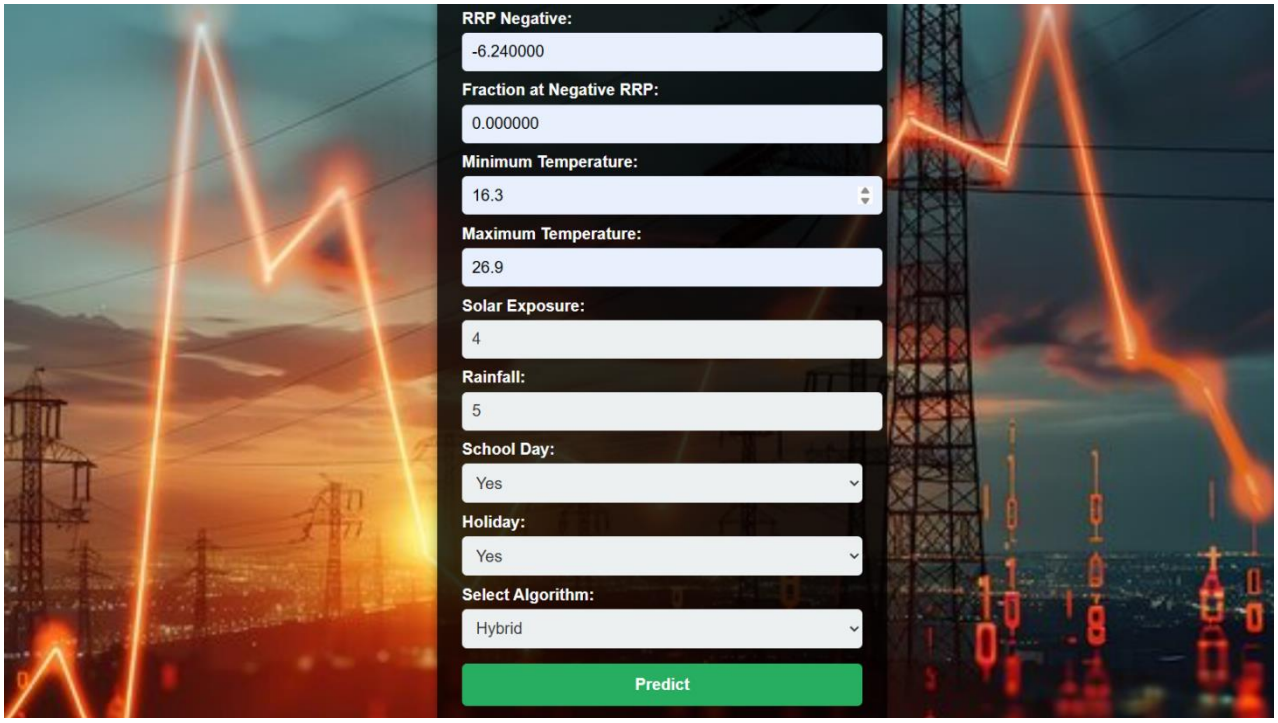

Fig 5.3.2 input Page

Fig 5.3.3 Prediction page

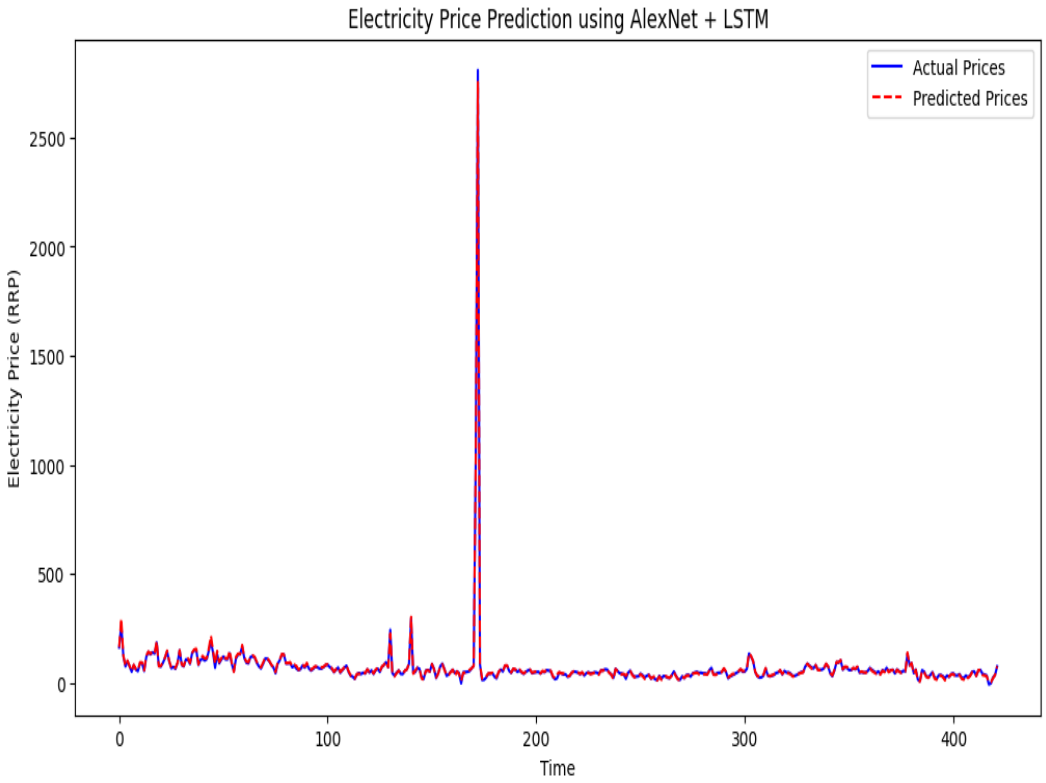

Fig 5.3.4 Graphs Page

The graph illustrates the performance of the hybrid AlexNet + LSTM model in predicting electricity prices over time. The x-axis represents time intervals, while the y-axis denotes the electricity price measured in RRP (Regional Reference Price). The plot compares actual prices (in blue solid line) with predicted prices (in red dashed line).It is evident that the hybrid model effectively tracks the actual price trends, including sharp price spikes and fluctuations, showcasing its capability to handle both stable and volatile pricing periods. Notably, around the mid-point of the graph, a significant price spike is accurately predicted by the model, indicating strong responsiveness to sudden market changes. The close alignment between actual and predicted values throughout the time series demonstrates the model's high accuracy and robustness in forecasting.This visual validation supports the quantitative metrics, affirming that the AlexNet + LSTM hybrid approach captures both spatial and temporal patterns critical for electricity price prediction.

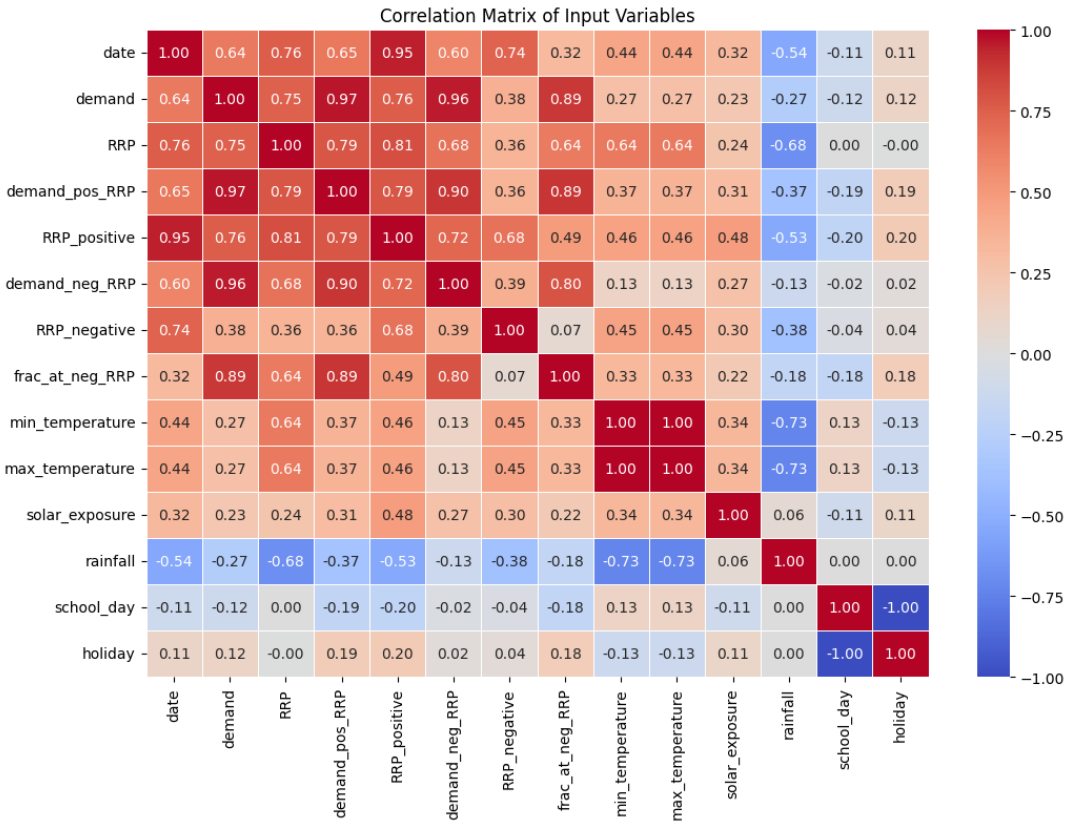

Fig 5.3.5 Correlation Matrix

The correlation matrix of input variables provides insights into the linear relationships between various features involved in electricity price prediction. From the matrix, it is evident that certain variables exhibit strong positive correlations. For instance, electricity demand shows a high positive correlation with demand_pos_RRP (0.97), RRP_positive (0.76), and RRP (0.75), indicating that as demand increases, electricity prices tend to rise as well. Similarly, demand_pos_RRP is strongly correlated with RRP_positive (0.90) and frac_at_neg_RRP (0.89), suggesting that these variables are closely aligned in capturing positive price trends and market dynamics. The variable RRP also correlates significantly with RRP_positive (0.79), reinforcing its dependency on upward price movements. On the contrary, negative correlations are observed with features like rainfall and temperature, particularly with RRP and RRP_positive, highlighting that weather conditions can inversely affect price levels. Variables such as school_day and holiday show weak or no significant correlation with the main price and demand indicators. Overall, this correlation matrix helps in identifying influential features and reducing multicollinearity when designing the predictive model.

6 CONCLUSION

The hybrid model based on AlexNet and LSTM proposed in this study is actually a breakthrough for electricity price forecasting since it could overcome the drawbacks associated with traditionally used methods such as ANN and RNN. Deep spatial features extracted from AlexNet with the sequential dependency of a model through LSTM let researchers analyze factors that influence electricity prices more comprehensively. That is, this hybrid combination captures both long- and short-term patterns in the data in a manner that will much improve prediction accuracy. Indeed, the hybrid model achieves a very impressive accuracy of 97.08% over traditional models, thus showing the strength and reliability in handling complex datasets that are involved in high-dimensional features and temporal dependencies. Further, the use of advanced measures, such as RMSE and MAE, will minimize the errors of this prediction, making that model highly effective to be applied in energy markets whose applications depend on the accuracy of the forecasted price. Success in the hybrid model is seen and actually emerges from the integration of multiple architectures designed to tackle complex real-world problems, more so in sectors like energy where accurate forecasts can yield some really good, large economic and operational benefits. This model is scalable and adaptable for future developments in the energy sector, including but not limited to, changes in policy and the addition of renewable energy sources making it a promising tool for long-term price prediction. The methodology applied in this study can even be extended to other domains where the time series and image data both form important components, thereby indicating flexibility and potential for future innovations in the model.

FUTURE WORK

The results of the electricity price prediction model using AlexNet + LSTM indicate promising performance, but there is still room for improvement and further exploration. Future work can focus on refining the model by incorporating additional influencing factors such as weather conditions, fuel prices, industrial demand, and renewable energy generation patterns. These factors play a crucial role in electricity price fluctuations and integrating them into the predictive model may enhance its accuracy and robustness.

The handling of extreme price spikes is another critical aspect of future research. The current model successfully captures price trends, but significant deviations in peak values suggest that it could be improved further. Developing anomaly detection mechanisms and integrating them into the forecasting pipeline could enhance the model's ability to handle sudden price surges. Methods such as using autoencoders for outlier detection or reinforcement learning-based prediction strategies may prove beneficial.

Moreover, expanding the dataset to include more historical data from different regions and timeframes can strengthen the generalizability of the model. A broader dataset will allow for better learning patterns and improve the model's adaptability to different market conditions. Additionally, real-time price forecasting using streaming data can be explored, enabling energy providers to make timely decisions. Implementing this model in an online learning framework, where it continuously updates with new data, can further enhance its accuracy over time.

CERTIFICATES

IEEE IDCIoT REVA UNIVERSITY Bengaluru, India

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- Transfer of Knowledge through contemporary curriculum and fostering faculty and student development.
- Create keen interest for research and innovation among students and faculty by understanding the needs of the society and industry.
- Skill development among diversity of students in technical domains and profession for development of systems and processes to meet the demands of the industry and research.
- Imbibing values and ethics in students for prospective and promising engineering profession and develop a sense of respect for all.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

After few years of graduation, the graduates of B. Tech (CSSE) will:

1. Demonstrate competencies in the Computer Science domain and Management with an ability to comprehend, analyze, design and create software systems for pursuing advanced studies in the areas of interest.
2. Evolve as entrepreneurs or be employed by acquiring required skill sets for developing computer systems and solutions in multi-disciplinary areas.
3. Exhibit progression and professional skill development in Computer programming and systems development with ethical attitude through life-long learning.

PROGRAM OUTCOMES (POs)

On successful completion of the Program, the graduates of B.Tech (CSSE) program will be able to:

1. Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

On successful completion of the Program, the graduates of B.Tech (CSSE) program will be able to:

PSO1 Employ Systems Approach to model the solutions for real life problems, design and develop software systems by applying Modern Tools.

PSO2 Develop solutions using novel algorithms in High Performance Computing and Data Science.

PSO3 Use emerging technologies for providing security and privacy to design, deploy and manage network systems.

COURSE OUTCOMES (COs)

After successful completion of this course, the students will be able to:

CO1. Create/Design algorithms and software to solve complex Computer Science and allied problems using appropriate tools and techniques following relevant standards, codes, policies, regulations and latest developments.

CO2. Consider Society, Health, Safety, Environment, Sustainability, Economics and project management in solving complex Computer Science and allied problems.

CO3. Perform individually or in a team besides communicating effectively in written, oral and graphical forms on Computer Science and allied systems or processes.

CO-PO and PSO Mapping Table:

Course Outcomes	Program Outcomes												Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	3	-	-	-	3	3	3	3
CO2	-	-	-	-	-	3	3	-	-	-	3	-	3	3	3
CO3	-	-	-	-	-	-	-	-	3	3	-	-	3	3	3
Average	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Level of correlation of the course	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

Correlation Level: 3 - High 2 - Medium 1 - Low